

BRAC University
Department of Computer Science and Engineering

Examination: Quiz 03
Duration: 20 Minutes

Semester : Spring 25
Full Marks: 15

CSE420: Compiler Design

Name:	ID:	Section:06
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1. Draw the parse tree and evaluate the attribute values for the input string: $8 + 12 * 100$

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L -> {value = 0; Stack.push(value);value = 4;}E{Stack.push(value);}
E -> E1 + T{value = T.val;value = Stack.pop(); Stack.push(value);} {E.val = E1.val + T.val;}
E -> T {value = value + T.val;E.val = T.val;}
T -> T1 * F { Stack.pop(); T.val = T1.val * F.val; value = T.val + T1.val - F.val;
    Stack.push(value);}
T -> F {T.val = F.val; value = value + F.val; Stack.push(value);}
F -> digit {F.val = digit.lexval; value = digit.lexval * digit.lexval; Stack.push(value);}
```

The token for the integer numbers will be digit. Show the dependency graph for the parse tree.